Antarctica Freeboard Calculation Based on ICEsat-2 and Tide Gauge

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Note: The text of this report was independently written by the author and subsequently polished with the assistance of ChatGPT.

1. Abstract

Sea ice plays a crucial role in climate regulation, ocean circulation, and ecosystem maintenance. It is commonly quantified using two key metrics: sea ice extent and thickness. Freeboard, an important indicator of sea ice thickness, represents the portion of ice above the water surface. This report estimates freeboard in Antarctica by combining ICESat-2 and tide gauge data.

For the tide gauge data, time windows with values clearly outside the normal range were first removed. To address data jumps, an offset correction was applied based on the assumption that sea level did not change dramatically within two months before and after data jump. Anomalous spikes and outliers were then removed to obtain a smooth time series. Finally, the tide gauge data in local coordinates were converted to ellipsoid heights by incorporating benchmark and tide gauge zero information. For the ICESat-2 ATL07 data, strong beams within 3 km of the tide gauge were selected according to the satellite's flight direction. After quality filtering, the ice surface height in ellipsoid coordinates was calculated. By combining the ellipsoid heights from both datasets, the freeboard variation trend was derived.

The experimental results show that although freeboard exhibits a periodic variation, the presence of negative values challenges the applicability of this method. These negative values are likely attributable to the poor measurement quality of the tide gauge, which stems from a lack of maintenance and regular calibration, as well as the incomplete spatial and temporal matching between the tide gauge and ICESat-2 data.

2. Introduction

Monitoring long-term and seasonal variations in sea ice and freeboard contributes to a better understanding of climate change processes. Traditional monitoring methods mainly include drilling holes and ice mass-balance buoys; however, in situ

experiments are labor-intensive, time-consuming, and particularly difficult to carry out in Antarctica [1-3]. Although remote sensing methods—such as spaceborne radar and lidar altimetry, passive microwave radiometry, synthetic aperture radar—have been widely applied, their use remains challenging because most satellite orbits do not adequately cover polar regions [4-7]. ICESat-2's orbit, in contrast, can reach high latitudes, covering most of Greenland and Antarctica, thereby making the remote sensing retrieval of polar freeboard feasible.

In addition, Xie et al. [8] proposed a method for calculating freeboard by combining GNSS-IR with a tide gauge. This approach not only improves the accuracy of freeboard monitoring but also compensates for the limitation of ICESat-2 in nearshore applications. Inspired by that study, this report attempts to use ICESat-2 as a substitute for GNSS-IR to obtain ice surface heights and then combine them with tide gauge data to compute freeboard.

3. Data

3.1 Tide Gauge

Tide gauges are generally capable of providing sea surface height information. However, in scenarios where polar regions are covered by ice sheets, radar-type tide gauges receive reflected signals from the ice surface. Under such circumstances, only Ocean Bottom Pressure (OBP)-type tide gauges can provide sea surface height instead of ice surface. Table 1 presents detailed information on the locations, sensor types, and benchmarks of tide gauges around Antarctica. Figure 1 illustrates the spatial distribution of these 8 stations. From table 1, we can see that only ROTHERA, ROBT and SCOT have benchmark information among 8 stations. Nevertheless, the datum for ROTHERA provided by Permanent Service for Mean Sea Level (PSMSL) is not complete. Therefore, within the entire Antarctic region, the stations that ultimately meet the requirements for freeboard calculation are limited to ROBT and SCOT.

Table 1 Tide Gauge Station on Antarctica.

Note: IOC is Intergovernmental Oceanographic Commission(https://www.ioc-sealevelmonitoring.org, last access on 04/21/2026); LINZ is Land Information New Zealand (https://www.linz.govt.nz/, last access on 04/21/2026).

Station	Lat, lon	Sensor type	Benchmark	Source
ROTHERA	-67.5667, -68.1333	OBP + Radar	Yes	IOC

ARGENTINE ISLANDS	-65.25 , -64.27	OBP + Radar	NO	IOC
PUERTO SOBERANIA	-62.4793, -59.6629	OBP + Radar	NO	IOC
OHIG3	-63.3203, -57.8987	OBP + Radar	NO	IOC
SYOWA	-69.0078, 39.5703	OBP	NO	IOC
DUMONT D'URVILLE	-66.6519, 140.0017	OBP	NO	IOC
ROBT	-77.0333, 163.1833	OBP	Yes	LINZ
SCOT	-77.8500, 166.7667	OBP	Yes	LINZ

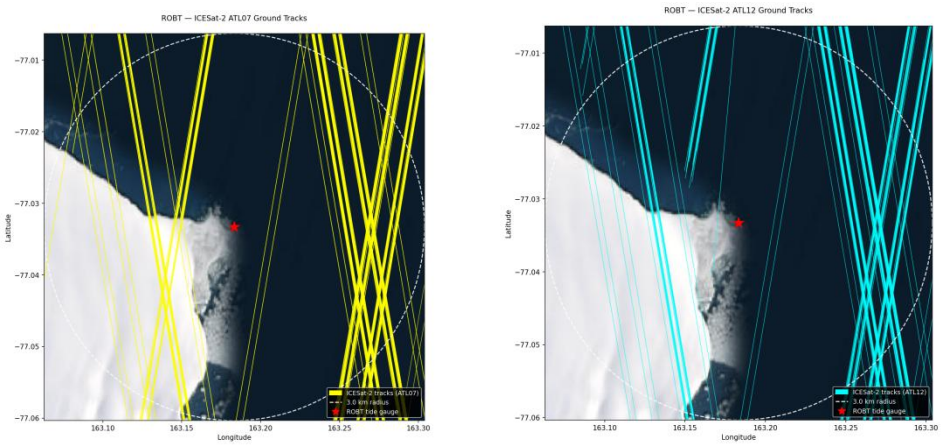
Figure 1 Antarctica Tide Gauge Distribution.

3.2 ICEsat-2

ICESat-2 is a Light Detection and Ranging (LiDAR) altimetry satellite that was launched in 2018. It employs a green laser with a wavelength of 532 nanometers and splits this laser beam into three pairs of beams for detection. ICESat-2 can generate a variety of product data from ATL03 to ATL22. In this experiment, we primarily utilized ATL07 and ATL12. ATL07 provides along-track surface height and type (e.g.,

snow-covered ice, open water) for the ice-covered seas of the northern and southern hemispheres (https://nsidc.org/data). ATL12 provides along-track sea surface height at variable length scales over cloud-free ocean regions. Estimates of height distribution, significant wave height, and sea state bias are also provided (https://nsidc.org/data).

This report used ATL07 to calculate ice surface height, also employed the ocean surface height provided by ATL12 to compare it with tide gauge. Figure 2 and figure 3 show ICEsat-2 ATL07 and ATL12 tracks aound ROBT stations, respectively.



(a) (b)

Figure 2 ICEsat-2 ATL07 and ATL12 tracks around ROBT station.

4. Experiments

4.1 Tide gauge Process

The ROBT site has been conducting calibration work on the tide gauge zero almost every year since 1991, while the SCOT site has been doing so since 2016. However, it is regrettable that the measurement data from these two sites exhibit bad stability. Table 2 illustrates the data missing conditions at both sites.

From table 2, we can see that in March 2021, tide gauge failure at the ROBT site results in abnormal data, which persisted until November 6. At the SCOT site, missing data or data recorded incorrectly due to instrument failures during four periods: from February 26, 2019, to October 30, 2019; from January 25, 2020, to May 12, 2020; from March 12, 2021, to November 18, 2021; and from March 9, 2022, to October 31, 2022. Moreover, since 2023, the tide gauge at the SCOT site has stopped recording data. For dealing with data missing problem, this experiment has decided to mask the

data missing period. Figure 3 and 4 show tide gauge sequences before and after masking the periods with missing data at ROBT and SCOT stations. After removing time windows with abnormal data, the ranges of tide gauge data seem more reasonable.

Table 2 The status of tide gauge data at ROBT and SCOT stations

Station	Date or Period	Reasons	Consequences
ROBT	2019	Operation issues	Unreliable Zero Calibration
	Feb/03/2019	Tide gauge failure	Data jump
	2020	Covid 19	No Zero Calibration
	Mar/17/2021 - Nov/6/2021	Tide gauge failure	Data error
SCOT	Feb/26/2019 - Oct/30/2019	Tide gauge failure	Data error
	Jan/25/2020 - May/12/2020	Tide gauge failure	Data error
	Mar/12/2021 - Nov/18/2021	Tide gauge failure	Data error
	Mar/09/2022 - Oct/31/2022	Tide gauge failure	Missing Data
	2023 - 2025	Tide gauge failure	Missing Data

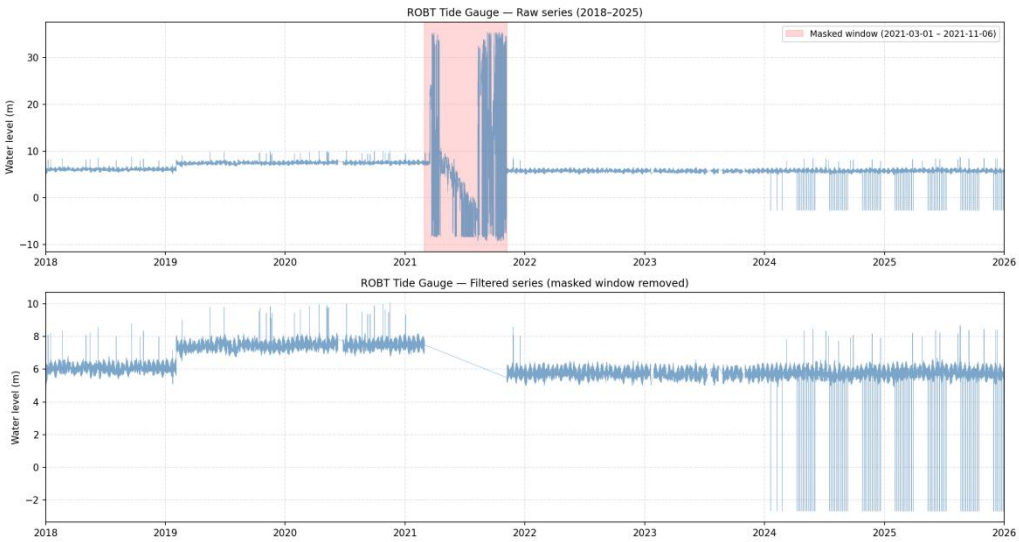


Figure 3 Tide gauge sequences before and after masking the periods with missing data at ROBT

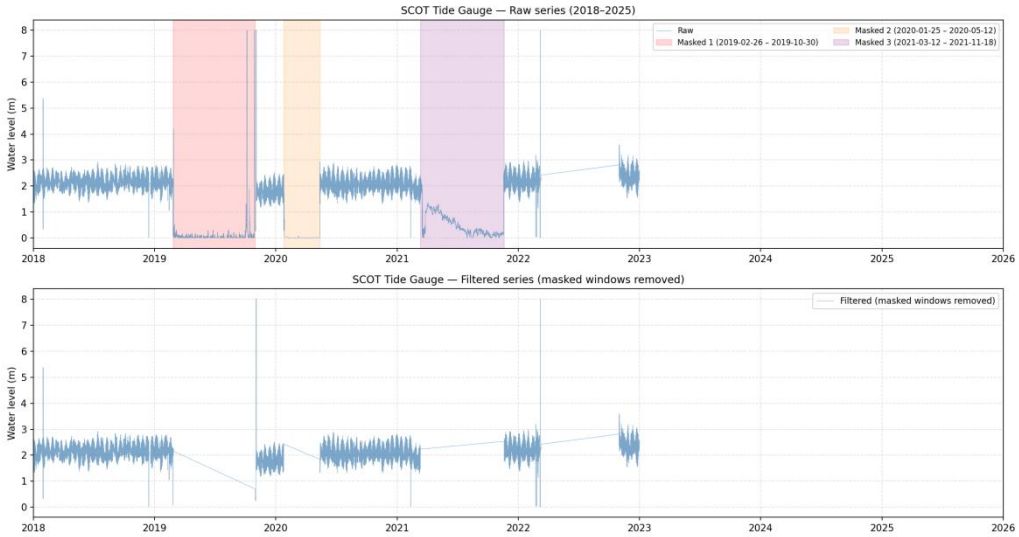


Figure 4 Tide gauge sequences before and after masking the periods with missing data at SCOT

In addition to the issue of missing data, shown as figure 3, the tide gauge data from the ROBT site also demonstrated a notable shift of approximately 1.5 meters on February 3, 2019. Furthermore, due to the impact of the COVID-19 pandemic, calibration activities were not carried out in 2020 [8]. In response to the data jump, given the limited availability of calibration data, and considering the tide gauge zero does not exhibit a consistent drift pattern, so we gave an assumotion: the sea level will not undergo significant changes within the two-month period preceding and following the data shift [8]. Subsequently, by utilizing the daily mean sea level data, which has been de-tided within 60 days before and after the data jump, the corresponding offset is calculated, which is then treated as an additional calibration reference point. The upper subfigure in figure 4 shows the data jump phenomenon, while the lower subfigure in figure 4 illustrates the tide gauge sequences jump removal after calibration.

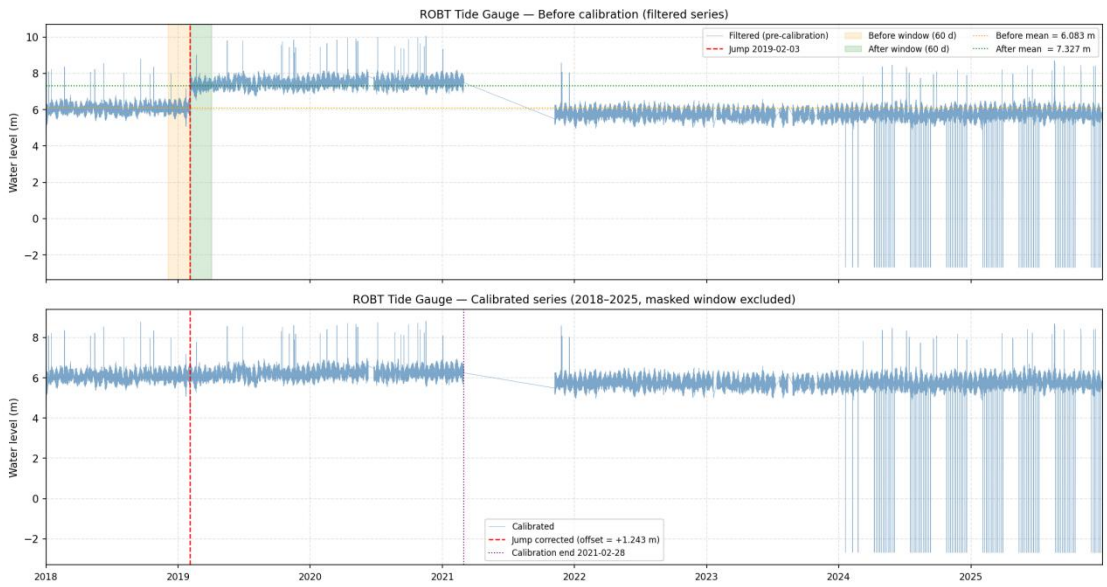


Figure 4 Data jump removal of tide gauge data for ROBT

In addition, this experiment also found a number of spiky anomalies in the tide gauge data. To address this issue, this experiment established the following criterion for outlier detection: if the observed value from the tide gauge at a specific sampling instant deviates from the daily mean by more than 2 meters, that data point is classified as an outlier and subsequently excluded from further analysis. Figure 5 and 6 show the outlier removal results at ROBT and SCOT, respectively.

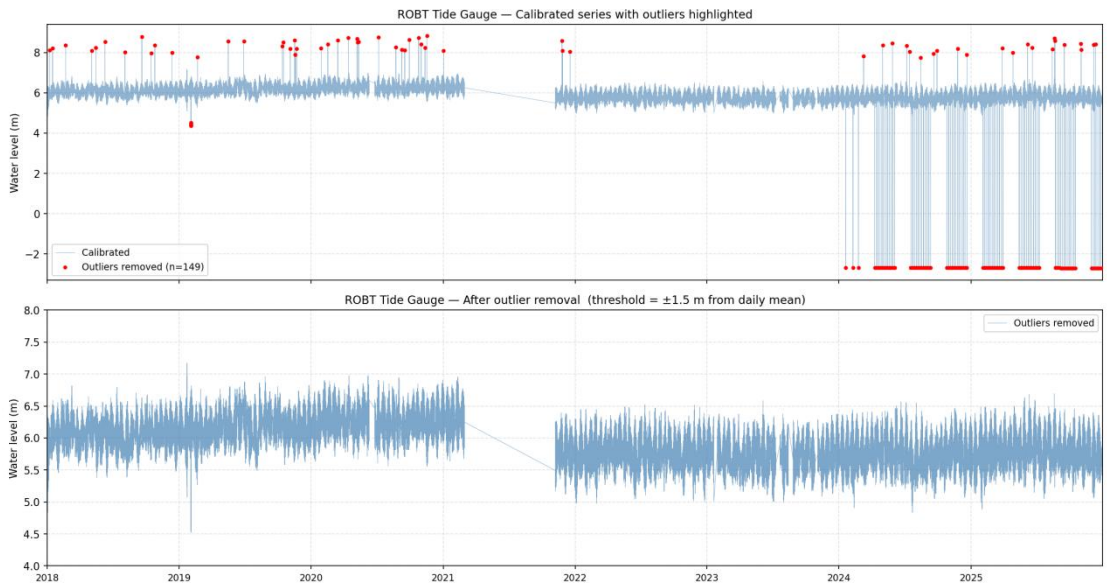


Figure 5 Outliers removal at ROBT

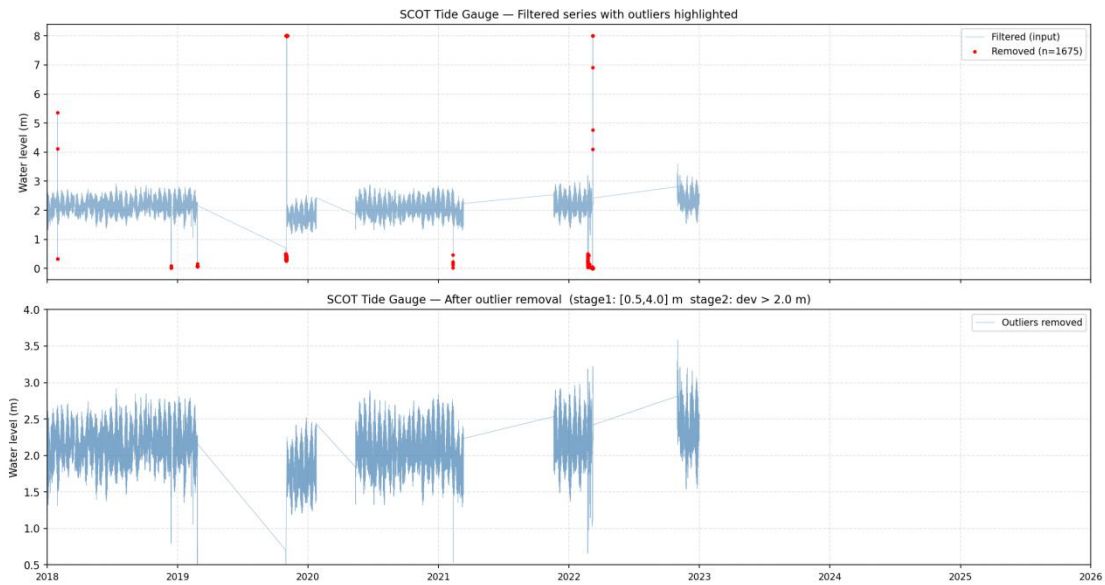


Figure 6 Outliers removal at SCOT

4.2 ICEsat-2 Process

The processing of ICEsat-2 data primarily consists of the following steps: strong beam selection, quality filtering, and elevation calculation.

1. Strong Beam Selection

The laser pulses from ICESat-2 generate three pairs of footprints on land surface, thereby producing six distinct ground tracks. The leftmost track designated as 1L (GT1L) and the rightmost as 3R (GT3R). Specifically, ground tracks 1L and 1R constitute the first pair, 2L and 2R the second pair, and 3L and 3R the third pair. However, each beam pair exhibits varying emission energies, categorized as weak and strong beams, as illustrated in the figure 7.

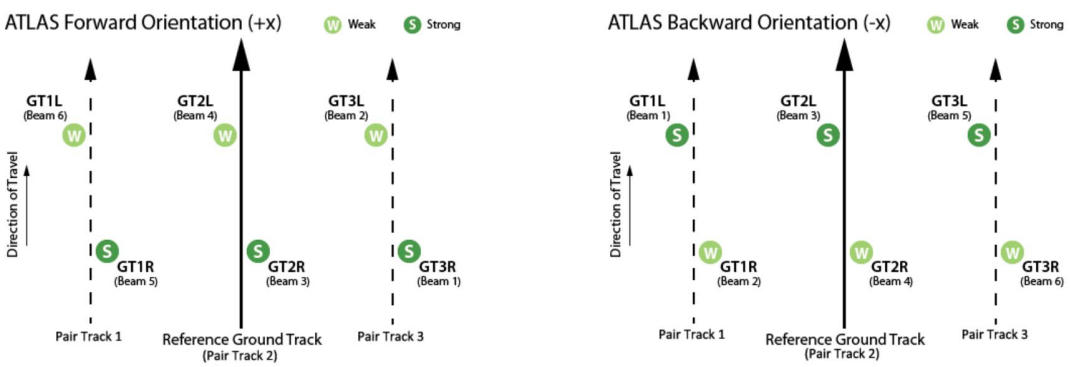


Figure 7 Strong and weak beams in the forward and backward direction
(Cited from: https://nsidc.org/data/atl07/versions/7)

In this experiment, we selected strong beams to extract data . As depicted in Figure 7, when the satellite is oriented in the forward direction, the strong beams are identified as GT1R, GT2R, and GT3R; conversely, in the backward direction, GT1L, GT2L, and GT3L serve as the strong beams. This is:

sc_orient = 1 (Forward): GT1R, GT2R, GT3R are strong;

sc_orient = 0 (Backward): GT1L, GT2L, GT3L are strong.

2. Data Filtering

After selecting the strong beams, it is necessary to further identify ice surface sampling points with good data quality. This experiment adopted the filtering criteria outlined in the following table as the basis for selection.

Table 3 Quality filtering for ICEsat-2

Field	Filter
`qa_granule_pass_fail`	== 0 (pass)
`height_segment_quality`	== 1 (good)
height_segment_type`	-1 = off_pointing, 0 = cloud, 1 = ice, 2~9 = lead
`height_segment_ssh_flag`	0 = ice surface, 1 = sea surface (lead) , -1 = fill value

Therefore, after meeting quality requirements, the sea ice could be detected by this equation:

sea_ice = height_segment_ssh_flag == 0 and height_segment_type == 1

3. Height Calculation

The data structure of ICEsat-2 includes two fields, named height_segment_height and height_segment_mss. Specifically, height_segment_height represents the height value relative to the Mean Sea Surface (MSS); whereas height_segment_mss reflects the height of the MSS relative to the WGS84 ellipsoid.

Therefore, to calculate the height value of the sea surface in the ellipsoidal coordinate system, the following equation should be employed. Figure 8 shows the ellipsoidal height from ICEsat-2 ATL07 ar ROBT and SCOT.

height_ellpsoid = height_segment_mss + height_segment_height

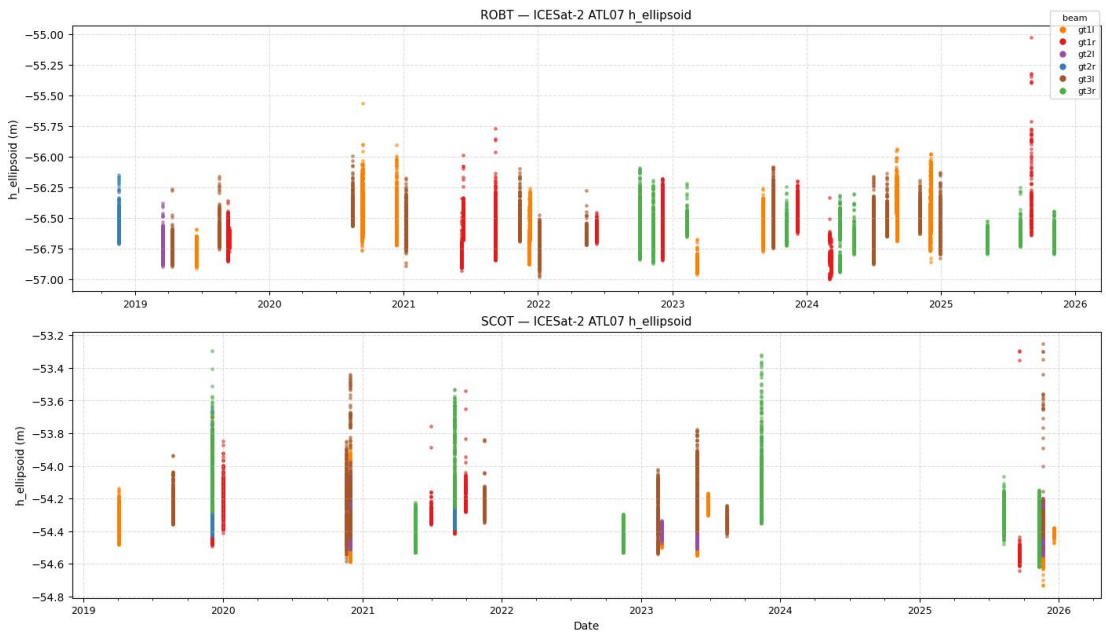


Figure 8 ICEsat-2 ATL07 ellipsoidal height at ROBT and SCOT

4.3 Unify to Ellipsoid

To calculate the freeboard, it is necessary to unify the heights of the ice surface and the water surface into the same coordinate system. Section 4.2 has successfully obtained the ellipsoidal height data of ICEsat-2. Hence, Section 4.3 will talk how to convert the height from the local coordinate measured by tide gauges to the ellipsoid coordinate system.

Both the ROBT and SCOT stations employ OBP-tide gauges, which record pressure changes at 5-minute intervals. The tide gauge zero undergoes leveling surveys almost annually with the assistance of GNSS receivers installed on sea ice during the southern hemisphere's spring. The location information of the benchmarks at the ROBT and SCOT stations is detailed in Figure 9. Furthermore, Table 4 presents the relevant leveling survey information for the ROBT and SCOT stations.

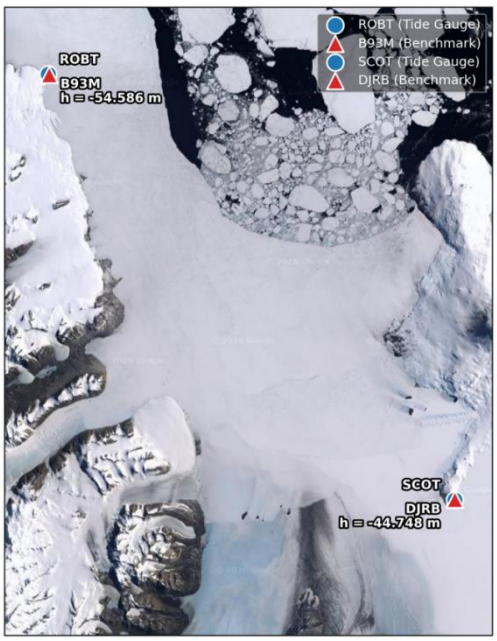


Figure 9 Benchmark and tide gauge zero at ROBT and SCOT

Table 4 Summary of tide gauge zero below benchmark B93M or DJRB

Station	Calibration Date	Value (m)
ROBT	November 2018	8.215
	October 2019	9.534 - unreliable
	November 2021	7.766
	November 2022	7.726
	November 2023	7.691
	November 2024	7.768
	November 2025	7.820
SCOT	November 2018	11.573
	November 2019	11.232
	November 2020	11.409
	November 2021	11.533
	November 2022	11.678
	November 2023	11.698

As shown in figure 10, after determining the information of the benchmark and the tide gauge zero, the formula for calculating the tide gauge height can be derived as:

$$\text{TG_ellipsoid} = \text{Benchmark_ellipsoid} - (\,|\text{tide_gauge_zero}| - \text{TG_local})$$
$$= \text{Benchmark_ellipsoid} + \text{tide_gauge_zero} + \text{TG_local}$$

Figure 11 illustrates the ellipsoidal heights of tide gauges ROBT and SCOT after coordinate transformation. We can see from this figure that during 2018 to 2025, the mean ellipsoidal heights of ROBT and SCOT are -56.561m and -54.128m.

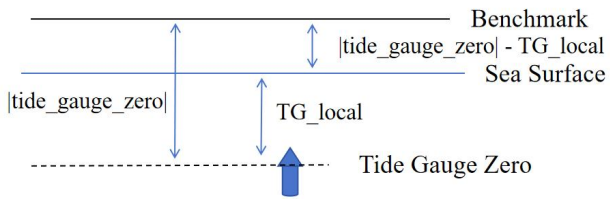


Figure 10 Ellipsoidal height calculation for tide gauge

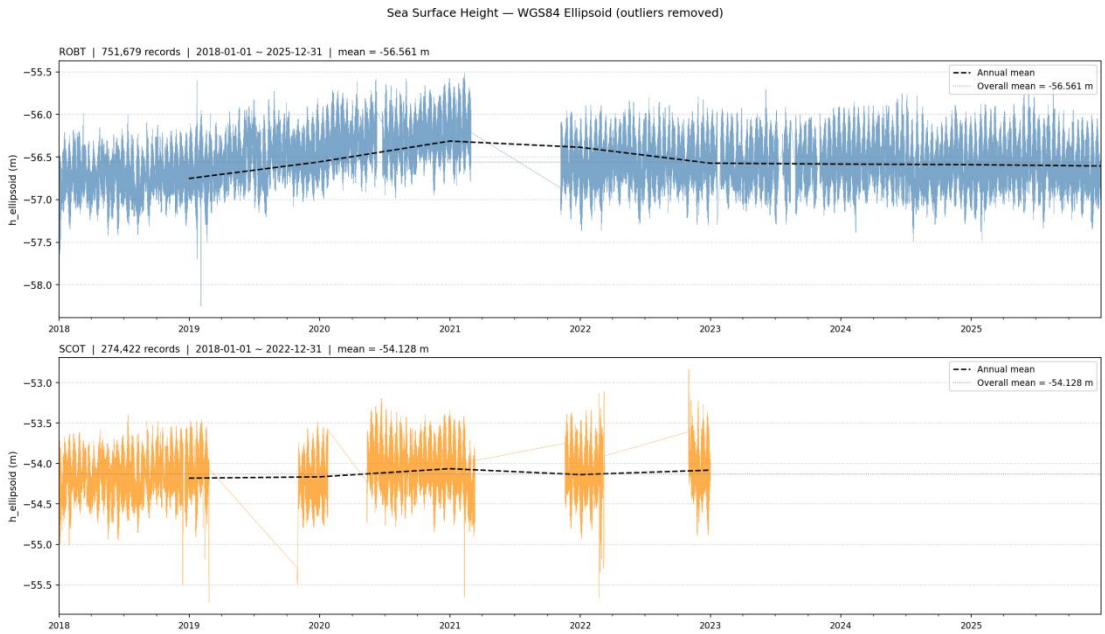


Figure 11 Ellipsoidal heights of tide gauges ROBT and SCOT after coordinate transformation.

4.4 Freeboard Calculation

After both the ice surface height results measured by ICEsat-2 ATL07 and the water surface height results measured by the tide gauge are unified into the ellipsoid coordinate system, the freeboard on the day of ICEsat-2's overpass can be calculated by subtracting the mean height measured by the tide gauge on that day from the mean height of all sampling points within a 3 km radius centered around the tide gauge as measured by ICEsat-2, shown as following equation.

$$\text{Freeboard} = \text{mean(ICEsat-2 height)} - \text{mean(tide gauge ellipsoid)}$$

Figure 12 give estimated freeboard change trends at ROBT and SCOT. ROBT site has 43 freeboard measurements, whereas the SCOT site has only 5 results. This is because, within a research area of 3-km radius, ICESat-2 traveled only 23 times in total around SCOT. Moreover, the tide gauge at SCOT stopped recording data in 2022, yet ICESat-2 still recorded 12 overflights after 2022. Even worse, the SCOT data contain several time gaps during which ICESat-2 had another 6 overflights. Therefore, only five valid measurements finally remained at SCOT, which is unfavourable for subsequent data analysis. Therefore, this report focuses primarily on the ROBT site.

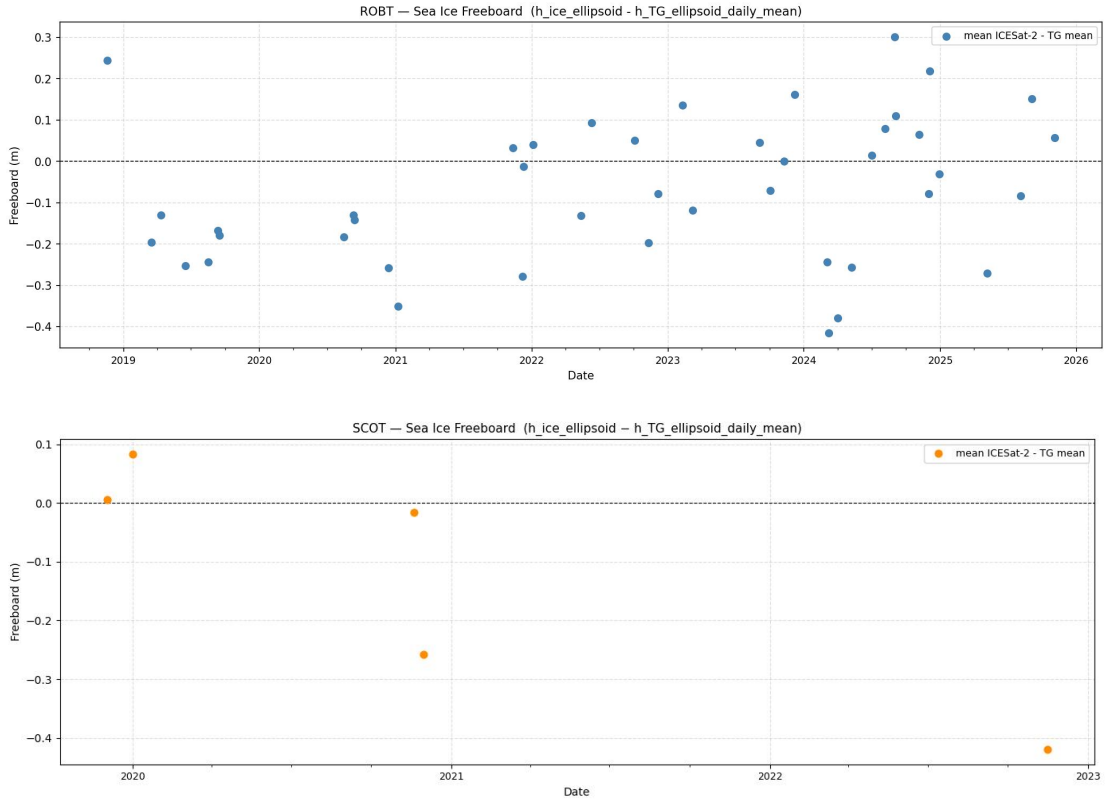


Figure 12 Freeboard change trends at ROBT and SCOT

The results in figure 12 show that the freeboard values generally lie in the range of –0.4 m to 0.3 m. However, freeboard is defined as the height of the ice above the water surface and should therefore be positive. Nevertheless, a large number of negative values appear in the results. It should be noted that this comparison uses the absolute ellipsoidal heights measured by ICESat-2 and the tide gauge, rather than comparing detrended or mean-removed variations. The negative values indicates that the absolute heights measured by ICESat-2 are systematically lower than those from the tide gauge, suggesting the presence of a systematic bias between the two datasets.

This bias threatens feasibility of estimating freeboard by combining tide gauge and ICESat-2 data.

5. Discussion

Section 4.4 reveals that ICESat-2 height measurements are systematically lower than tide gauge observations. In the Discussion, this discrepancy will be determined whether it stems from computational errors or reflects a true systematic bias, and potential reasons will be analyzed.

5.1 freeboard periodic variation

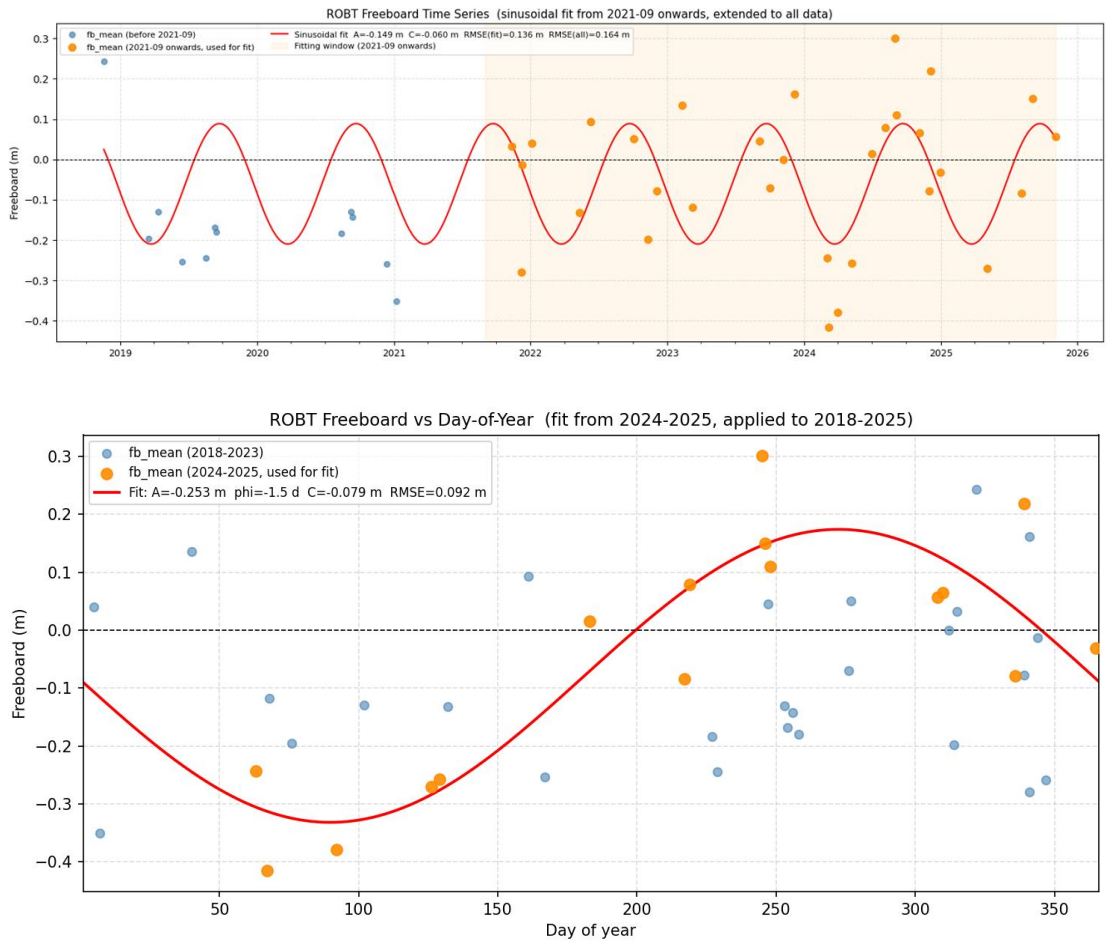


Figure 13 The periodic variation of freeboard

In Antarctica, freeboard height typically follows a annual cycle [8]. The upper panel of figure 13 presents the fitted periodic variation from early 2019 to late 2025. Despite the sparse data, which weakens the periodic signal, a seasonal trend is evident from early 2022 to late 2025: freeboard rises in the austral winter and falls in the austral summer. This suggests that, although the absolute height bias between ICESat-2 and

the tide gauge yields negative freeboard values, the trend still conforms to the annual cycle characteristic.

5.2 Comparison between tide gauge and lead

Leads (ice cracks) serve as a key proxy for sea surface height and are essential for estimating freeboard with ICESat-2. Comparison of lead-derived sea surface heights from ICESat-2 ATL07 with tide gauge measurements (see figure 14) reveals that, within 3 km of ROBT, only three leads were observed. The tide gauge recorded sea surface heights approximately 20–60 cm higher than those retrieved from ICESat-2. This further highlights a systematic bias between ICESat-2 and tide gauge estimates of absolute height in the ellipsoidal reference frame.

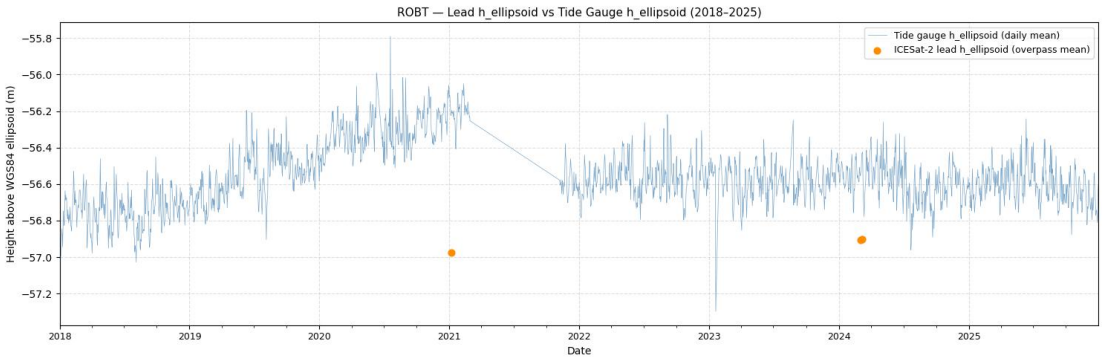


Figure 14 The periodic variation of freeboard

5.3 Comparison between tide gauge and ATL12

The ICESat-2 ATL12 product provides sea surface height data. A comparison between the heights derived from ATL12 and tide gauge observations in the ellipsoidal reference frame is presented in Figure 15. The figure shows that the sea surface heights from ATL12 are about 10–20 cm lower than those from the tide gauge.

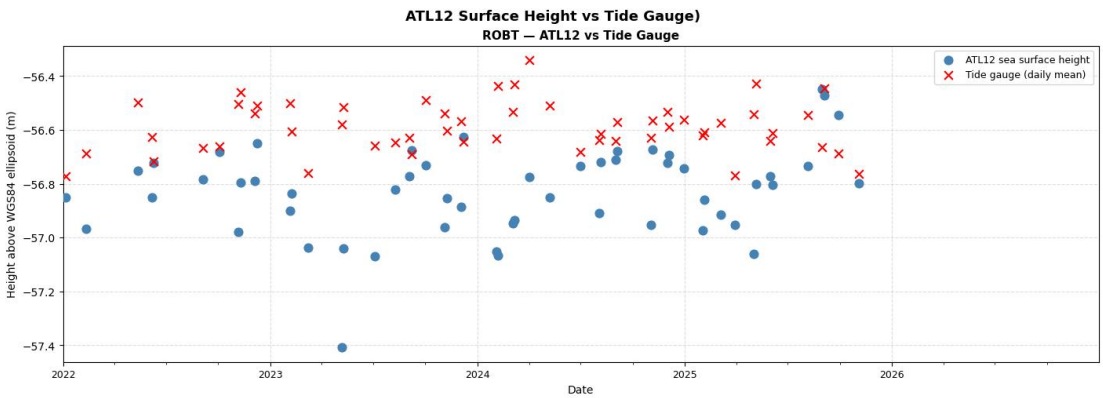


Figure 15 The periodic variation of freeboard

5.4 Reasons

Based on the above analysis, the systematically lower sea surface and ice surface heights derived from ICESat-2, compared to tide gauge observations, are not attributable to calculation errors but rather to systematic bias. Possible sources of this bias include：

(1) The lack of maintenance and calibration for the tide gauge has resulted in data drift.

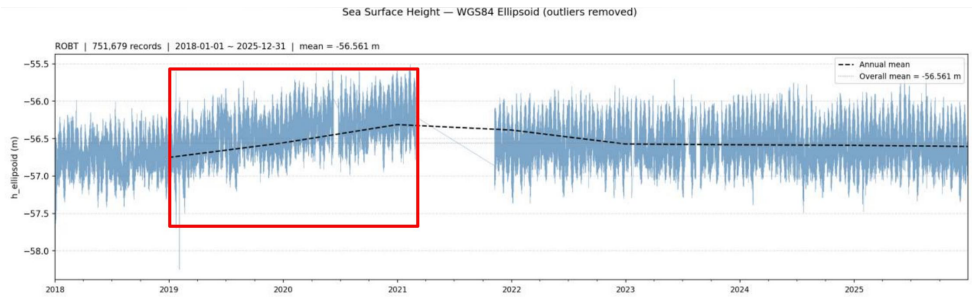


Figure 16 Data drift in tide gauge

As shown as table 2, the zero calibration was unreliable in 2019 and there was no zero calibration in 2020, resulting in that the whole two-year tide gauge data have not been calibrated at all. This maked tide gauge data drift around 50 cm during just two years, which then increase the uncertainty and decrease the quality of tide gauge data.

(2) Tide gauge measurements represent daily mean sea levels, while ICESat-2 data are averages over specific time windows and spatial areas, making them more sensitive to tidal variations.

(3) The ICESat-2 sampling area does not perfectly align with the tide gauge location, introducing a spatial mismatch.

(4) Cloud cover and harsh weather in the Antarctic winter reduce the precision of ICESat-2's atmospheric delay corrections, causing range overestimation and height underestimation.

(5) The reference ellipsoid used by ICESat-2 may differ from that of the tide gauge benchmark.

5.5 limitation

Although this study identifies systematic errors in the method of estimating Antarctic freeboard by combining ICESat-2 and tide gauge data, we still need to acknowledge that data sparsity and low quality may also affect the conclusions.

First, only two sites in the polar region possess complete benchmark information, and harsh weather conditions cause instrument instability, leading to data drift or temporal

gaps, which undoubtedly pose difficulties for analysis. Second, ICESat-2 overflights are inherently sparse, and the loss of tide gauge data further reduces the number of valid observations—yielding only 5 at the SCOT site—thus severely limiting the validation capacity of this method.

6. Conclusion

This study estimates freeboard at ROBT and SCOT by integrating ICESat-2 observations with tide gauge measurements. For the tide gauge data, periods with values falling outside the normal range were first excluded, and then an offset adjustment was performed to correct data jumps under the assumption that sea level underwent no significant change during the two months before and after each jump. After removing anomalous spikes and outliers, a smoothed time series was obtained. Finally, the locally referenced tide gauge data were converted to ellipsoidal heights using the benchmark and tide gauge zero information. Regarding the ICESat-2 ATL07 data, strong beams located within 3 km of the tide gauge were firstly selected. Following quality filtering, ice surface heights in the ellipsoidal reference frame were then calculated. By combining the ellipsoidal heights from both datasets, the trend of freeboard variation was derived.

The experimental results indicate that, although freeboard shows some periodic changes, the large number of negative values raises doubts about the method's reliability. These negative values are likely attributable to the poor measurement quality of the tide gauge, which stems from a lack of maintenance and regular calibration, as well as the incomplete spatial and temporal matching between the tide gauge and ICESat-2 data.

7. References

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